2- order PDE

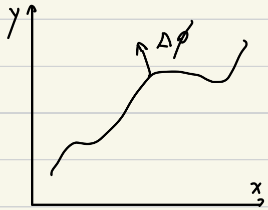
$$a(x,y)\,u_{xx} + b(x,y)\,u_{xy} + c(x,y)\,u_{yy} = f(x,y,u,u_x,u_y)$$

Discriminant $\Delta = b^2 - 4ac$

Characteristic line.

$T: \phi(x,y) = 0$

$\nabla\phi = (\phi_x, \phi_y)$



fix $(x,y) = (x_0, y_0)$

$\eta = \phi(x,y) \qquad \xi = \psi(x,y) \qquad$ where $\begin{vmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{vmatrix} \neq 0$

$u_x = u_\xi \xi_x + u_\eta \cdot \eta_x \qquad u_y = u_\xi \xi_y + u_\eta \eta_y$

$u_{xx} = u_{\xi x}\xi_x + \cdots + u_\eta \eta_{xx}$

$\quad = \xi_x^2 u_{\xi\xi} + \cdots + \eta_{xx} u_\eta$

$u_{xy} = u_{\xi y}\xi_x + \cdots + u_\eta \eta_{xy}$

$\quad = \xi_x \xi_y u_{\xi\xi} + \cdots + \eta_{xy} u_\eta$

$u_{yy} = u_{\xi y}\xi_y + \cdots + u_\eta \eta_{yy}$

$\quad = \xi_y^2 u_{\xi\xi} + \cdots + \eta_{yy} u_\eta$

coefficient of $u_{\eta\eta}$:

$A\eta_x^2 + B\eta_x\eta_y + C\eta_y^2$

coefficient of $u_{\xi\xi}$:

$A\xi_x^2 + B\xi_x\xi_y + C\xi_y^2$

法向二阶导?

if $A\eta_x^2 + B\eta_x\eta_y + C\eta_y^2 = 0$. write $\phi(x,y) = y - f(x)$

$$A f'^2 - B f' + C = 0$$

$$\Delta = B^2 - 4AC$$

$\Delta < 0$ Elliptic

$\Delta = 0$ parabolic

$\Delta > 0$ hyperbolic

方程在曲线法向 (ϕ_x, ϕ_y) 上退化

$$x^2 y'' + a x y' + b y = 0 \qquad t = \ln|x| \qquad y(x) = u(t)$$

$$y' = \frac{dy}{dx} = \frac{du}{dt}\frac{dt}{dx} = \dot{u} \cdot \frac{1}{x}$$

$$y'' = \frac{dy'}{dx} = \frac{d\,\dot{u}\cdot\frac{1}{x}}{dx} = \frac{\frac{1}{x}d\dot{u} + \dot{u}\cdot(-\frac{1}{x^2})dx}{dx} = \frac{1}{x^2}\ddot{u} + (-\frac{1}{x^2})\dot{u}$$

$$u'' + (a-1)u' + bu = 0$$

Heat equation.

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

Initial condition $u(x,0) = f(x)$

PBCs $u(0/L, t) = 0$

$$u(x,t) = X(x)\,T(t)$$

$$X(x)T'(t) = X''(x)T(t)$$

$$\frac{T'(t)}{T(t)} = \frac{X''(x)}{X(x)} = \lambda$$